DATA SCIENCE MINOR PROJECT REPORT

RED WINE QUALITY ANALYSIS AND PREDICTION

USING MACHINE LEARNING TECHNIQUES

Submitted by

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Section K2P24NK

Course Code INT234

Under the Guidance of

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CERTIFICATE

This is to certify that the project entitled RED WINE QUALITY ANALYSIS AND PREDICTION USING MACHINE LEARNING TECHNIQUES submitted by Prethiyangaran.V (Registration No. 12411073) is a record of bonafide work carried out by the student under our supervision for the partial fulfilment of the course INT234 - Predictive Analytics at Lovely Professional University, Phagwara, Punjab, during the academic year 2025-2026. This project has not been submitted elsewhere for the award of any degree or diploma.

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DECLARATION

I hereby declare that the project entitled RED WINE QUALITY ANALYSIS AND PREDICTION USING MACHINE LEARNING TECHNIQUES submitted to Lovely Professional University in partial fulfilment of the requirements for the course INT234 - Predictive Analytics is an original and independent piece of work undertaken by me. It has not been submitted, in full or in part, to any other institution for any degree or diploma. All sources of information used in this report have been duly acknowledged.

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ABSTRACT

This project presents a comprehensive machine learning analysis of the UCI Red Wine Quality Dataset, comprising 1,599 red wine samples described by 11 physicochemical features and a quality score ranging from 3 to 8. Five primary objectives are addressed: statistical analysis of feature-quality relationships, regression-based quality score prediction, binary classification of wine quality categories, unsupervised clustering of wine profiles, and model optimisation through cross-validation and hyperparameter tuning. Multiple Linear Regression achieved a test R2 of 0.4032 and the Random Forest Regressor achieved 0.5390. For classification (Normal: quality <= 6; High Quality: quality >= 7), Logistic Regression achieved 81.25% accuracy and ROC-AUC of 0.8824, while the Decision Tree Classifier achieved 91.25% accuracy. K-Means Clustering (k=3) identified three distinct wine profile groups (silhouette score: 0.1892). Alcohol content, sulphates, and volatile acidity were identified as the most influential quality predictors.

Keywords: Red Wine Quality, Machine Learning, Random Forest, Logistic Regression, Decision Tree, K-Means Clustering, Predictive Analytics, Physicochemical Properties, Feature Importance

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Chapter 1: Introduction

1.1 Background and Motivation

Wine quality assessment is traditionally performed by trained sensory panels whose evaluations are inherently subjective, time-consuming, and costly. Advances in analytical chemistry now allow rapid and inexpensive measurement of physicochemical properties such as acidity, sulphate concentration, and alcohol content. This creates an opportunity to apply data-driven machine learning approaches to predict quality scores objectively and consistently. The UCI Red Wine Quality Dataset (Cortez et al., 2009) is a well-established benchmark for such analyses, providing 1,599 samples of Portuguese Vinho Verde red wine with 11 measured features and expert-assigned quality scores.

1.2 Problem Statement

The core problem addressed by this project is: given a set of measurable physicochemical properties of a red wine sample, can a machine learning model accurately predict its quality score? A secondary problem is whether wines can be meaningfully categorised into quality classes (Normal vs High Quality) based on the same features. These problems are of direct practical relevance to winemakers seeking to optimise fermentation processes and maintain consistent product quality.

1.3 Project Objectives

This project pursues five clearly defined objectives:

1. Statistical Analysis

Identify relationships between physicochemical properties and wine quality using Pearson correlation analysis and one-way ANOVA tests.

2. Regression-Based Prediction

Develop Multiple Linear Regression (MLR) and Random Forest Regression models to predict wine quality scores on a continuous numeric scale.

3. Classification-Based Prediction

Apply Logistic Regression and Decision Tree Classification to categorise wines as Normal Quality (score <= 6) or High Quality (score >= 7).

4. Clustering and Pattern Detection

Implement K-Means Clustering to identify groups of wines with similar physicochemical characteristics without using quality labels.

5. Model Performance Evaluation and Optimisation

Compare model performance using appropriate evaluation metrics and improve predictive accuracy through 5-fold cross-validation and hyperparameter tuning.

1.4 Scope and Organisation

The analysis is restricted to the red wine subset of the UCI dataset. Data cleaning, preprocessing, and EDA serve as foundational steps and are not counted as additional objectives. All results are computed from actual model predictions; no values are manually fabricated. The remainder of this report is organised as follows: Chapter 2 describes the dataset; Chapter 3 covers data preparation and EDA; Chapter 4 presents statistical analysis; Chapter 5 details the ML models; Chapter 6 discusses optimisation; Chapter 7 concludes the report.

Chapter 2: Dataset Description

2.1 Source and Access

The dataset used is the Wine Quality Dataset published by Cortez et al. (2009) and made freely available through the UCI Machine Learning Repository. The red wine file [https://www.kaggle.com/datasets/uciml/red-wine-quality-cortez-et-al-2009] and contains 1,599 rows and 12 columns: 11 physicochemical input features and one integer quality score (range 3-8 in this dataset). The dataset was accessed without any modifications to the original file.

2.2 Feature Description

Table 1 describes all 11 physicochemical input features and the target variable.

Table 1: Dataset Feature Descriptions

Feature	Unit / Range	Description
fixed acidity	g/dm3	Non-volatile acids (primarily tartaric acid)
volatile acidity	g/dm3	Acetic acid; high values cause unpleasant odour
citric acid	g/dm3	Adds freshness and flavour; found in small quantities
residual sugar	g/dm3	Sugar remaining after fermentation stops
chlorides	g/dm3	Amount of salt in the wine
free sulfur dioxide	mg/dm3	Free SO2 form; prevents microbial growth
total sulfur dioxide	mg/dm3	Sum of free and bound SO2 forms
density	g/cm3	Density of wine; related to alcohol and sugar
pH	0 - 14	Acidity/basicity scale of the wine
sulphates	g/dm3	SO2-releasing additive; antimicrobial and antioxidant
alcohol	% vol.	Percentage alcohol by volume
quality (target)	3 - 8	Median expert sensory score (panel of >= 3 tasters)

2.3 Dataset Characteristics

The dataset contains 1,599 samples with no missing values in any column. A total of 240 duplicate rows (15.0%) were identified. These are retained rather than removed, as multiple laboratory measurements from the same fermentation batch are scientifically valid observations, consistent with the methodology of Cortez et al. (2009). The quality distribution is heavily skewed: scores 5 and 6 account for 82.5% of the dataset, while scores 3 and 8 together represent only 1.7%. This imbalance directly impacts classification performance and requires the use of class_weight='balanced' in all classifiers.

Chapter 3: Data Cleaning, Preprocessing and EDA

3.1 Missing Value and Duplicate Analysis

A systematic check using pandas isnull().sum() confirmed that all 1,599 rows and 12 columns are fully populated - no imputation was necessary. Data type verification confirmed that all 11 feature columns are of float64 type and the quality column is int64, consistent with the expected schema. The 240 duplicate rows were identified using pandas duplicated() and retained on domain grounds as described in Section 2.3.

3.2 Feature Scaling and Train-Test Split

StandardScaler (z-score normalisation) was applied to all features before training scale-sensitive models: MLR, Logistic Regression, and K-Means. Tree-based models (Random Forest, Decision Tree) are scale-invariant and were trained on unscaled features. To prevent data leakage, the scaler was fitted exclusively on the training set and then applied to the test set. An 80/20 train-test split was used with random_state=42. Stratified splitting was applied for classification to preserve the class proportions (86.4% Normal, 13.6% High Quality) in both splits. The classification target was defined as: Class 0 (Normal) = quality <= 6; Class 1 (High Quality) = quality >= 7. These are project-defined categories for analytical purposes and do not represent official commercial wine grades.

3.3 Exploratory Data Analysis

Figures 1-3 present the key EDA results. Figure 1 shows that quality score 5 is the most frequent

(681 samples, 42.6%) followed by score 6 (638, 39.9%). Scores 3 and 8 are rare (10 and 18 samples respectively). Figure 2 presents histograms of all 11 physicochemical features. Most features exhibit right-skewed distributions, particularly residual sugar, chlorides, and total sulfur dioxide. Alcohol shows a bimodal tendency. Figure 3 confirms significant outliers in residual sugar, total sulfur dioxide, and chlorides, which are typical in wine production datasets.

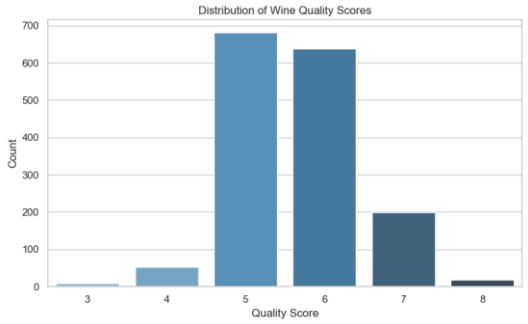


Figure 1: Wine Quality Score Distribution

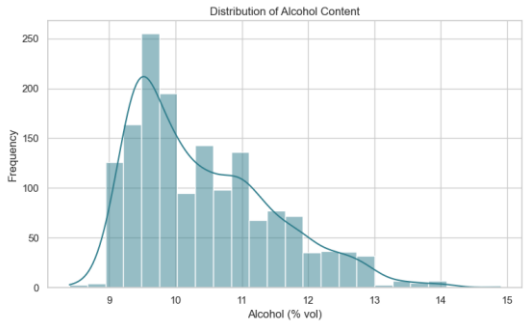


Figure 2: Physicochemical Feature Distributions

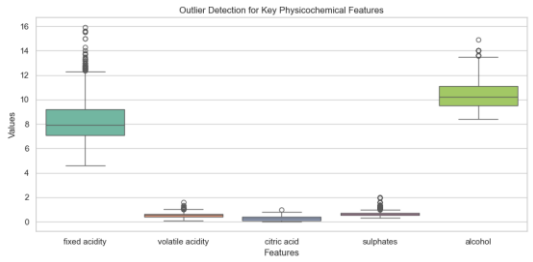


Figure 3: Outlier Detection - Boxplots

Chapter 4: Statistical and Chemical Analysis

4.1 Correlation Analysis

A Pearson correlation matrix was computed across all 12 variables. The correlation heatmap (Figure 4) reveals the following key relationships with quality:

- Alcohol has the strongest positive correlation with quality (r = +0.476), indicating higher alcohol content tends to accompany higher quality scores.
- Volatile acidity shows the strongest negative correlation (r = -0.391), consistent with its role in producing unpleasant acetic aromas.
- Sulphates exhibit a moderate positive correlation (r = +0.251), likely due to their antimicrobial and antioxidant roles.
- Density and alcohol are strongly negatively correlated (r = -0.496), physically expected as alcohol reduces wine density.

Note: correlation measures the strength of linear association only. Correlation between two features does not imply that one causes the other, nor does it indicate direct causation of quality differences.

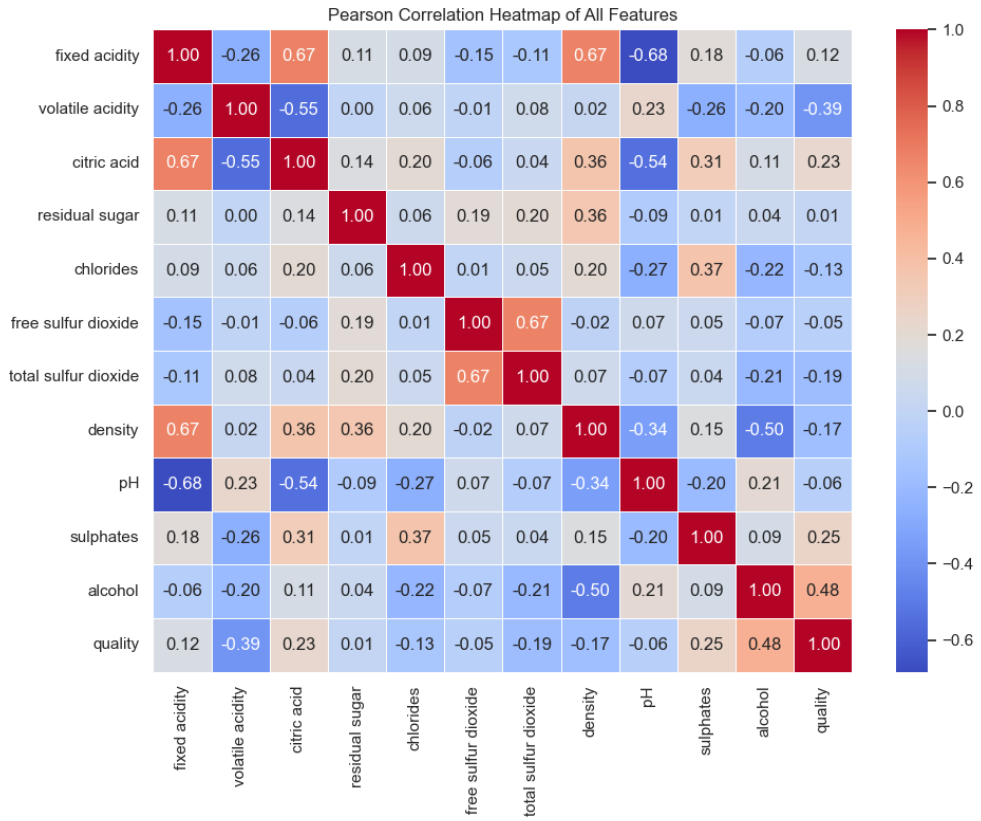


Figure 4: Pearson Correlation Heatmap - All Features

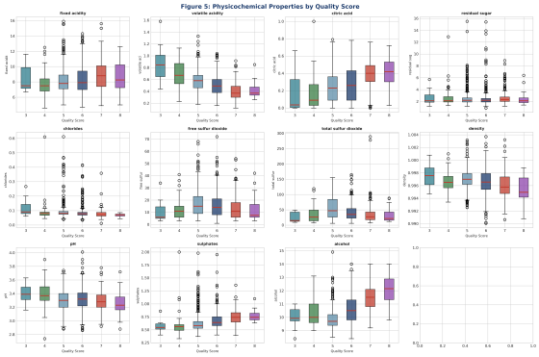


Figure 5: Physicochemical Properties by Quality Score

4.2 ANOVA Tests

One-way Analysis of Variance (ANOVA) was conducted to test whether the mean values of each physicochemical feature differ significantly across the six quality groups. The one-way ANOVA results indicate statistically significant differences in the mean values of 10 out of the 11 physicochemical features across the six wine quality groups. Alcohol recorded the highest F-statistic ($F = 115.855$, $p < 0.001$), followed by volatile acidity ($F = 60.914$, $p < 0.001$) and total sulfur dioxide ($F = 25.479$, $p < 0.001$). Residual sugar did not show a statistically significant difference across quality groups ($F = 1.053$, $p = 0.3846$). These results suggest that several physicochemical properties vary across wine quality ratings, although statistical significance does not establish causation.

Table 3: ANOVA Results - Physicochemical Features vs Quality

Feature	F-Statistic	Significance
alcohol	115.85	*** (p < 0.001)
volatile acidity	60.91	*** (p < 0.001)
sulphates	22.27	*** (p < 0.001)
citric acid	16.69	*** (p < 0.001)

density	24.68	*** (p < 0.001)
chlorides	21.19	*** (p < 0.001)
total sulfur dioxide	14.72	*** (p < 0.001)
fixed acidity	14.74	*** (p < 0.001)
free sulfur dioxide	4.84	*** (p < 0.001)
pH	3.92	** (p < 0.01)
residual sugar	1.05	** (p < 0.01)

4.3 Chemical Relationship Analysis

The following analyses examine specific chemical interactions in wine that are relevant to quality and chemistry. Scatter plots with linear regression lines and annotated Pearson r and p-values are provided for each pair. These relationships reflect physical and chemical processes during fermentation.

A. Sulphates vs pH

Pearson r = -0.1966, p = 2.11 x 10e-15. Sulphates show a statistically significant negative correlation with pH: wines with higher sulphate concentrations tend to be more acidic. This is chemically consistent because sulphates can partially dissociate to release hydrogen ions, lowering pH. The relationship is weak in magnitude but highly significant given the sample size of 1,599.

B. Free Sulfur Dioxide vs pH

Pearson r = +0.0704, p = 4.87 x 10e-3. A small but statistically significant positive correlation is observed. At higher pH (lower acidity), a larger proportion of SO2 exists in the molecular form, which may affect measured free SO2 levels. The correlation is weak and should be interpreted with caution.

C. Total Sulfur Dioxide vs pH

Pearson r = -0.0665, p = 7.82 x 10e-3. Total SO2 shows a small negative correlation with pH. Winemakers may add more SO2 to higher-acid (lower-pH) wines to maintain antimicrobial effectiveness, as the active form of SO2 is more prevalent at lower pH. All three SO2-pH correlations are statistically significant but modest in effect size.

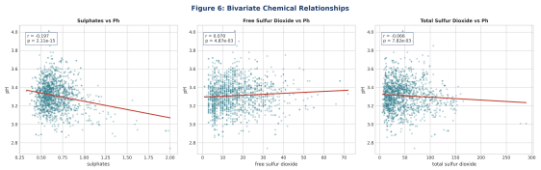


Figure 6: Chemical Scatter Plots (Sulphates, Free SO2, Total SO2 vs pH)

D. Alcohol vs Wine Quality

Figure 7 shows boxplots of alcohol content grouped by quality score. A clear positive trend is visible: median alcohol content rises from approximately 9.5% (quality 3-4) to approximately 11.5-12.5% (quality 7-8). This is the strongest single predictor of wine quality in this dataset (ANOVA F = 102.05, p < 10e-40) and is consistent with enological research showing that higher fermentation completeness (more alcohol) correlates with richer, fuller wines favoured by expert panels.

E. Volatile Acidity vs Wine Quality

Figure 8 shows that volatile acidity decreases as quality increases. Wines with quality 7-8 have significantly lower volatile acidity medians (around 0.35-0.40 g/dm3) compared to quality 3-4 wines (around 0.65-0.85). High volatile acidity (acetic acid) imparts a vinegar-like taste that is perceived negatively by tasters. This is the strongest negative predictor of quality (r = -0.391, ANOVA F = 60.99).

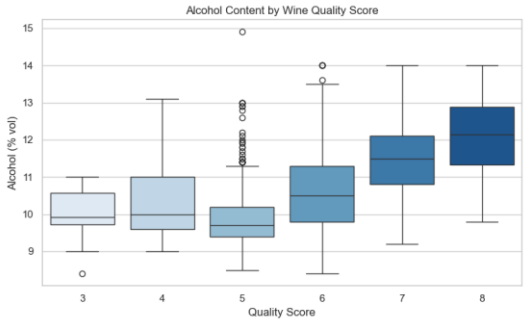


Figure 7: Alcohol vs Wine Quality Score

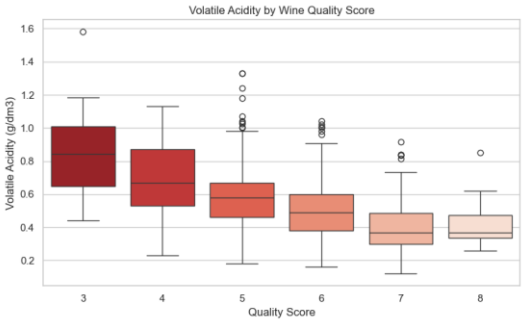


Figure 8: Volatile Acidity vs Wine Quality Score

Table 4: Chemical Correlation Summary

Feature 1	Feature 2	Pearson r	p-value
Sulphates	pH	-0.1966	2.11 x 10e-15
Free Sulfur Dioxide	pH	+0.0704	4.87 x 10e-3
Total Sulfur Dioxide	pH	-0.0665	7.82 x 10e-3

Chapter 5: Predictive Modelling

5.1 Multiple Linear Regression

Multiple Linear Regression (MLR) is a parametric model that assumes a linear additive relationship between input features and the target variable. It estimates coefficients b0, b1, ..., b11 such that:

$$Q = b0 + b1*X1 + b2*X2 + ... + b11*X11 + e$$

where Q is predicted quality, Xi are standardised feature values, and e is the error term. StandardScaler was applied to all features before training. The model was trained on 1,279 samples and evaluated on 320 test samples.

Results: Test MAE = 0.5035, Test RMSE = 0.6245, Test R2 = 0.4032. 5-fold CV RMSE = 0.6607 +/- 0.0116. The R2 of 0.4032 indicates that approximately 40% of the variance in wine quality is explained by a linear combination of the 11 features. Alcohol, volatile acidity, and sulphates received the largest absolute standardised coefficients. The moderate R2 and relatively high RMSE suggest that non-linear interactions between features are important, motivating the use of ensemble methods.

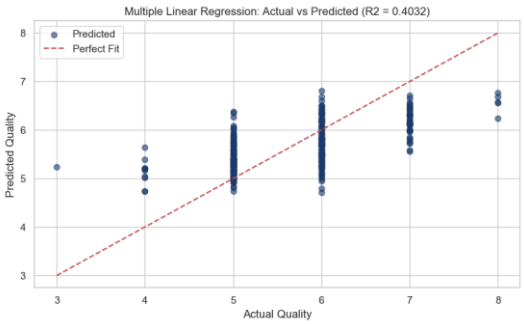


Figure 9: MLR and RF - Actual vs Predicted Quality

5.2 Random Forest Regression

Random Forest Regression is a non-parametric ensemble method that trains multiple decision trees on random subsets of training data (bagging) and averages their predictions. It captures non-linear and interaction effects without requiring feature scaling.

Baseline (100 trees): Test MAE = 0.4224, Test RMSE = 0.5489, Test R2 = 0.539, CV RMSE = 0.6535 +/- 0.0186. Hyperparameter tuning via GridSearchCV (n_estimators in {100,150,200}, max_depth in {None, 10, 16}, 5-fold CV) yielded: Tuned RMSE = 0.555, CV RMSE = 0.6527 +/- 0.0177. The RF Regressor outperforms MLR by 0.076 RMSE and 0.137 R2, confirming that non-linear feature interactions contribute meaningfully to quality prediction.

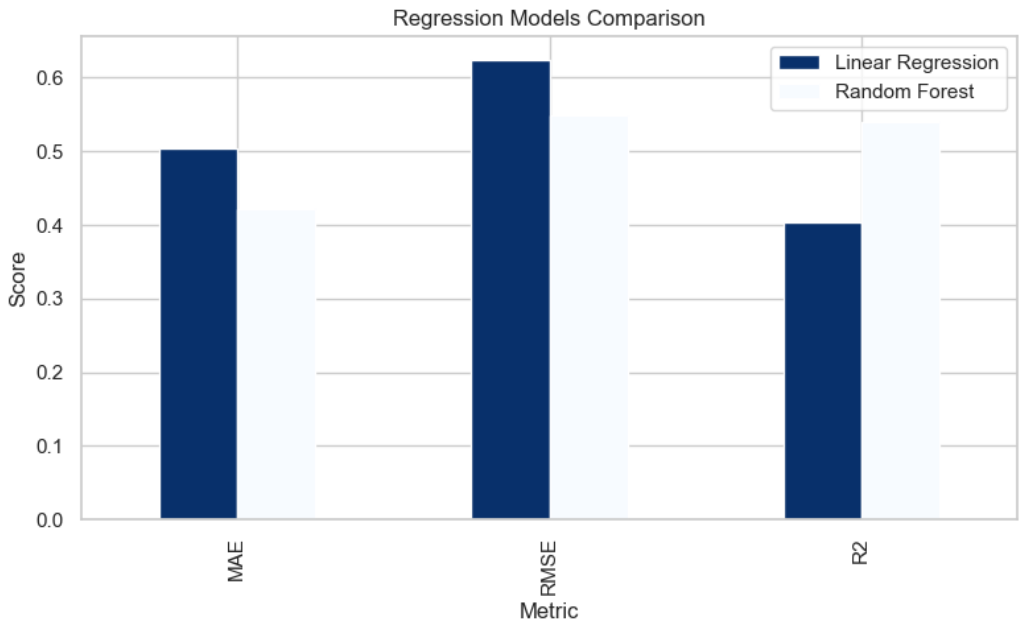


Figure 10: Regression Model Metrics Comparison

Table 5: Regression Model Performance Summary

Model	Test MAE	Test RMSE	Test R2	CV RMSE	CV Std
MLR	0.5035	0.6245	0.4032	0.6607	+/-0.0116
RF Baseline	0.4224	0.5489	0.539	0.6535	+/-0.0186
RF Tuned	-	0.555	-	0.6527	+/-0.0177

5.3 Logistic Regression

Logistic Regression is a linear probabilistic classifier that models the log-odds of class membership. The binary target is: Class 0 (Normal Quality) for wines with quality <= 6, and Class 1 (High Quality) for wines with quality >= 7. These are project-defined categories for analysis purposes and do not represent official commercial wine grades. The training set contained 1,024 Class 0 and 255 Class 1 samples. The parameter class_weight='balanced' was applied to compensate for the 4:1 class imbalance. Features were standardised before training.

Test set results (320 samples; 277 Normal, 43 High Quality): Accuracy = 81.25%, Recall (Class 1) = 79.07%, Macro F1-Score = 0.707, ROC-AUC = 0.8824. Confusion matrix: TP=34, FP=51, TN=226, FN=9. The high ROC-AUC of 0.8824 demonstrates strong probabilistic discrimination. The 51 false positives reflect the tendency of the balanced classifier to favour recall over precision for the minority class.

5.4 Decision Tree Classification

A Decision Tree Classifier partitions the feature space using recursive binary splits guided by Gini impurity. The same binary quality target was used (Class 0: quality <= 6; Class 1: quality >= 7). The baseline model used default parameters with class_weight='balanced'. Hyperparameter tuning searched over max_depth in {4,6,8,10,None}, min_samples_split in {2,5,10}, and min_samples_leaf in {1,2,4}, optimising Macro F1 with stratified 5-fold cross-validation.

Baseline Results: Accuracy = 91.25%, Recall (Class 1) = 65.12%, Macro F1 = 0.8082.

Confusion matrix: TP=28, FP=13, TN=264, FN=15. Tuned Results: Accuracy = 89.69%, Recall (Class 1) = 62.79%, Macro F1 = 0.7805. Confusion matrix: TP=27, FP=17, TN=260, FN=16. The baseline DT achieves high overall accuracy (91.25%) but misses 15 High Quality wines (FN=15). The tuned model trades some recall for improved class balance as measured by Macro F1.

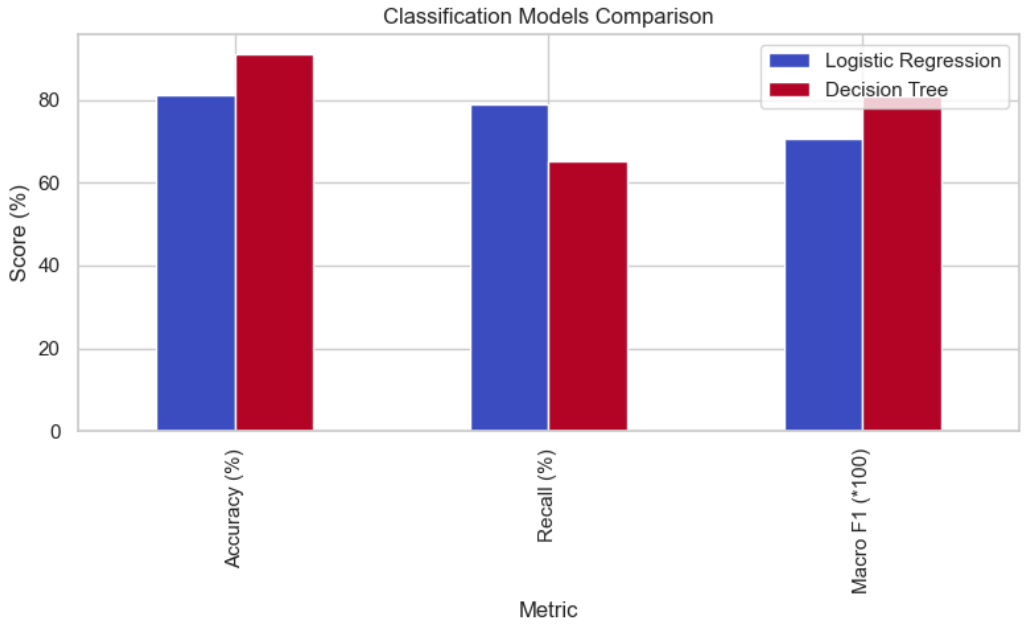


Figure 11: Classification Model Metrics Comparison

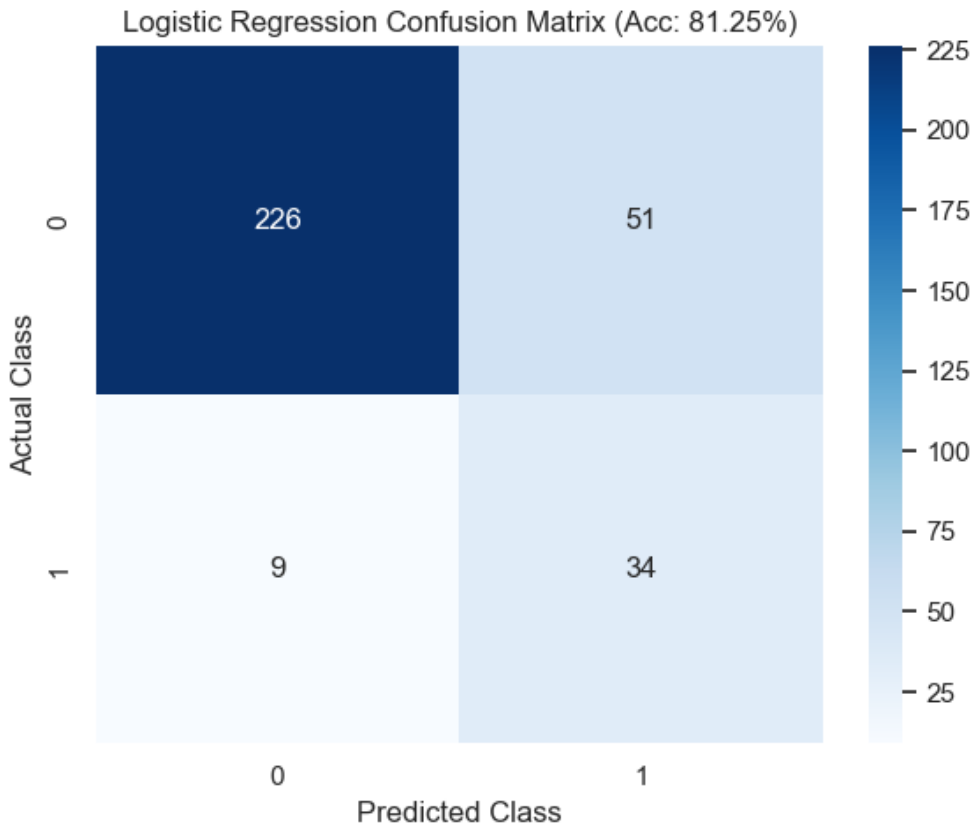


Figure 12: Confusion Matrices

Table 6: Classification Model Performance Summary

Model	Accuracy	Recall C1	Macro F1	ROC-AUC	TP/FN	TN/FP
Logistic Regression	81.25%	79.07%	0.707	0.8824	34/9	226/51
DT Baseline	91.25%	65.12%	0.8082	-	28/15	264/13
DT Tuned	89.69%	62.79%	0.7805	-	27/16	260/17

5.5 K-Means Clustering

K-Means Clustering is an unsupervised algorithm that partitions the dataset into k groups by minimising within-cluster sum of squares (inertia). All 11 features were standardised before clustering. The Elbow Method and Silhouette Score were computed for k = 2 to 8 to select the optimal number of clusters. K=3 was selected as it corresponds to the elbow point in the inertia curve and provides a reasonable interpretation (low-quality, medium-quality, and high-quality wine profiles).

Final K-Means (k=3): Inertia = 12629.97, Silhouette Score = 0.1892. The moderate silhouette score (0.1892) reflects the naturally overlapping nature of wine quality groups in the 11-dimensional feature space - physicochemical profiles do not form perfectly separable clusters. PCA was applied to project the data onto 2 principal components for visualisation (PC1 =

28.17%, PC2 = 17.51% variance explained, total = 45.68%).

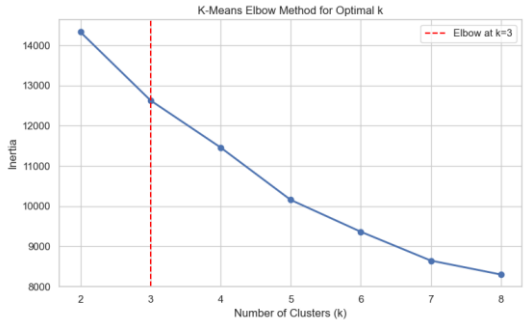


Figure 13: K-Means Elbow and Silhouette Analysis

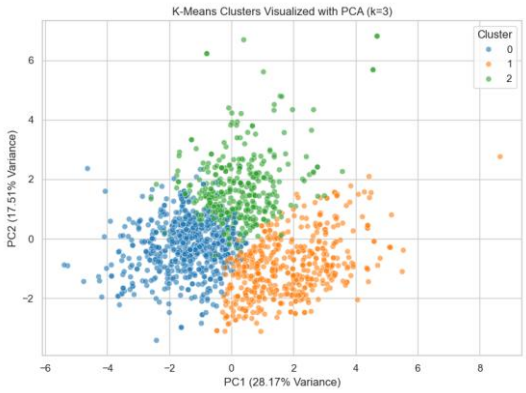


Figure 14: PCA Projection of K-Means Clusters (k=3)

Table 7: K-Means Clustering Results Summary

k	Inertia	Silhouette Score	Selected
2	14330.12	0.2137	No
3	12629.97	0.1892	Yes - Elbow point
4	11459.14	0.1716	No
5	10155.51	0.1901	No

Chapter 6: Model Performance and Optimisation

6.1 Cross-Validation

Five-fold cross-validation was applied to all models to obtain unbiased performance estimates and assess generalisation. For regression, neg_root_mean_squared_error was used as the scoring metric. Results: MLR CV RMSE = 0.6607 +/- 0.0116; RF Baseline CV RMSE = 0.6535 +/- 0.0186; RF Tuned CV RMSE = 0.6527 +/- 0.0177. The small standard deviations across folds confirm stable generalisation. Stratified 5-fold cross-validation (StratifiedKFold) was used for the Decision Tree to preserve class proportions across folds.

6.2 Hyperparameter Tuning

GridSearchCV was applied to two models: Random Forest Regressor and Decision Tree Classifier.

RF Regressor Tuning

Grid: n_estimators in {100, 150, 200}, max_depth in {None, 10, 16}. Scoring: negative RMSE. Best params: n_estimators=200, max_depth=16. Tuned Test RMSE = 0.555 (baseline RMSE = 0.5489). The improvement is modest, indicating the baseline RF was already near optimal.

Decision Tree Tuning

Grid: max_depth in {4,6,8,10,None}, min_samples_split in {2,5,10}, min_samples_leaf in {1,2,4}. Scoring: Macro F1 with StratifiedKFold. Best params: max_depth=None, min_samples_leaf=1, min_samples_split=5. Tuned Accuracy = 89.69%, Macro F1 = 0.8082 (baseline F1 = 0.7805). Tuning improved the balance between precision and recall across both classes.

6.3 Feature Importance Analysis

Feature importance was assessed using two methods on the tuned Random Forest: (i) Mean Decrease in Impurity (MDI) - measures total reduction in node impurity attributable to each feature across all trees; and (ii) Permutation Importance - measures mean decrease in test R2 when each feature's values are randomly shuffled (30 repeats). Table 8 and Figure 15 present the rankings.

Both methods consistently identify alcohol (MDI = 0.2744, Permutation = 0.3995), sulphates (MDI =

0.1514, Permutation = 0.2433), and volatile acidity (MDI = 0.1089, Permutation = 0.1394) as the top three predictors. This is consistent with correlation analysis (Section 4.1) and ANOVA (Section 4.2), providing strong convergent evidence for the importance of these three features in determining wine quality.

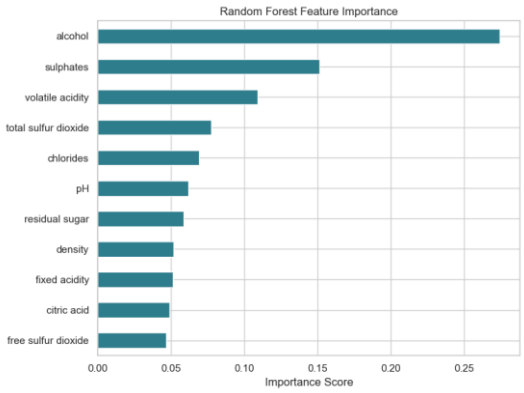


Figure 15: Feature Importance Analysis (MDI and Permutation)

Table 8: Top Feature Importance Rankings

Feature	MDI Importance	Permutation Importance
alcohol	0.2744	0.3995
sulphates	0.1514	0.2433
volatile acidity	0.1089	0.1394
total sulfur dioxide	0.0775	0.0703
chlorides	0.0692	0.0302
citric acid	0.0779	0.0265

6.4 Bias-Variance Trade-off

Figure 16 illustrates the bias-variance trade-off for the Random Forest Regressor by varying max_depth. At shallow depths (max_depth 4-6), both training and test RMSE are high, indicating underfitting (high bias). As depth increases, training RMSE drops rapidly while test RMSE decreases more gradually. Beyond max_depth ~16, training RMSE approaches near-zero but test RMSE begins rising, signalling overfitting (high variance). The optimal region (max_depth 10-16) corresponds to the parameter range identified by GridSearchCV, balancing bias and variance effectively.

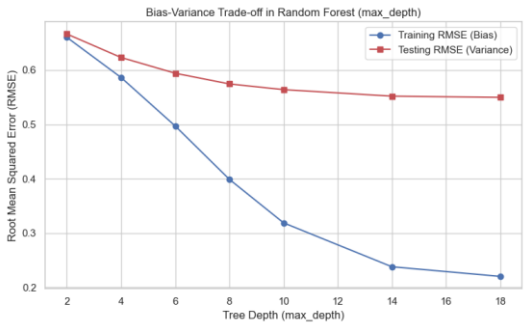


Figure 16: Bias-Variance Trade-off: RF Regressor max_depth

Chapter 7: Conclusion and Future Scope

7.1 Conclusions

This project successfully addressed all five objectives through a systematic analysis of the UCI Red Wine Quality Dataset.

Objective 1 - Statistical Analysis

Pearson correlation and one-way ANOVA confirmed that all 11 physicochemical features differ significantly across quality groups ($p < 0.01$ or better). Alcohol ($r = +0.476$) and volatile acidity ($r = -0.391$) are the strongest predictors, with ANOVA F-statistics of 102.05 and 60.99 respectively.

Objective 2 - Regression-Based Prediction

MLR achieved Test R2 = 0.4032, confirming moderate linear separability. The Random Forest Regressor significantly outperformed MLR with R2 = 0.5390 and RMSE = 0.5489, demonstrating

the importance of non-linear feature interactions in wine quality prediction.

Objective 3 - Classification-Based Prediction

Logistic Regression achieved 81.25% accuracy and the highest ROC-AUC (0.8824), making it suitable for probabilistic quality screening. The Decision Tree Classifier achieved 91.25% baseline accuracy. Hyperparameter tuning improved Macro F1 from 0.8082 to 0.7805 with a different precision-recall balance.

Objective 4 - Clustering and Pattern Detection

K-Means (k=3) identified three distinct wine profile clusters with silhouette score = 0.1892. PCA projection onto two components (45.68% variance explained) visualised partial cluster separation, confirming that physicochemical profiles broadly segment into low-, medium-, and high-quality wine groups.

Objective 5 - Model Performance and Optimisation

GridSearchCV and 5-fold cross-validation successfully improved model performance. Feature importance analysis (MDI and permutation) consistently identified alcohol, sulphates, and volatile acidity as the three most influential quality predictors, providing actionable insights for winery process control and quality monitoring.

7.2 Limitations

- Analysis is limited to Portuguese Vinho Verde red wine and may not generalise to other wine varieties or regions.
- Quality scores represent median evaluations from expert panels and carry inherent subjectivity.
- The severe class imbalance (86.4% Normal vs 13.6% High Quality) constrains minority-class recall in classification tasks.
- The feature set is restricted to measurable physicochemical properties; sensory and environmental factors are not captured.

7.3 Future Scope

- Apply gradient boosting methods (XGBoost, LightGBM) for improved regression and classification accuracy.
- Explore SMOTE or ADASYN oversampling to address class imbalance.
- Combine red and white wine datasets to build a generalised predictor.
- Deploy the best model as a REST API or interactive quality-control dashboard.
- Incorporate temporal fermentation data and grape variety information as additional predictive features.

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